

CONCEPT MAP : DIVIDE AND CONQUER → RECURRENCE RELATION

Divide and Conquer
→ Recurrence Relation
→ $T(n) = aT(n/b) + f(n)$
→ Master theorem

How each node contributes
in analyzing recursive
algorithm

Root (Level 0)

- * Problem of size n
- * Work done: $f(n)$

Internal Nodes

At level i , there are a^i subproblems

Size of each subproblem

$$= n/b^i$$

Work per node at level i :

$$\Theta(n/b^i)$$

Total work at level i :

$$a^i \cdot f(n/b^i)$$

Leaves (Base Case)

When subproblem size
reduces at constant (1)
Work per leaf: $\Theta(1)$

Total Work $T(n)$

Sum of work over all levels
Dominant contribution
decides the time complexity

Recurrence Relation
 $T(n) = aT(n/b) + f(n)$

$a \geq 1$: number of subproblems
 $a > 1$: factor by which the problem
size is divided
 $f(n)$: work done outside recursive
calls (worktime cost)

Master Theorem
Let $n \log_b a$ compare with $f(n)$

Case 1:
 $f(n) = O(n^{\log_b a - \epsilon})$
for some $\epsilon > 0$

$f(n)$ grows asymptotically
slower than $n^{\log_b a}$

$$T(n) = \Theta(n^{\log_b a})$$

Example:

$$T(n) = 2T(n/2) + n$$

$$\text{Here, } a=2, b=2, n^{\log_b a} = n$$

$$f(n) = n = O(n^{1-\epsilon}) \text{ for } \epsilon > 0$$

$$\Rightarrow T(n) = O(n)$$

Case 2:

$$f(n) = \Theta(n^{\log_b a} \log^k n)$$

for $k \geq 0$

$f(n)$ is within a logarithmic
factor of $n^{\log_b a}$

$$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

Example:

$$T(n) = 2T(n/2) + n$$

$$\text{Here } a=2, b=2, n^{\log_b a} = n$$

$$f(n) = n = \Theta(n^{\log_b a} \log^0 n)$$

$$\Rightarrow T(n) = \Theta(n \log n)$$

Case 3:

$$f(n) = \Omega(n^{\log_b a + \epsilon})$$

for some $\epsilon > 0$

$f(n)$ grows asymptotically
faster than $n^{\log_b a}$

$$T(n) = \Theta(f(n))$$

Example:

$$T(n) = 2T(n/2) + n^2$$

$$\text{Here, } a=2, b=2, n^{\log_b a} = n$$

$$f(n) = n^2 = \Omega(n^{\log_b a + \epsilon}) \text{ for } \epsilon=1$$

$$\Rightarrow T(n) = O(n^2)$$

n : Input size
 a : Number of subproblems
 b : Division factor
 $f(n)$: Non-recursive work
 $T(n)$: Time Complexity

CONCEPT MAP : DIVIDE AND CONQUER → PROBLEM DIVISION → SUBPROBLEMS → RECURRENT RELATION → MASTER THEOREM

1. DIVIDE AND CONQUER

A technique to solve problems by breaking them into smaller independent parts.

Solve and combine their solution

2. PROBLEM DIVISION

The problem of size n is divided into smaller subproblems each subproblem is size n/b .

3. SUBPROBLEMS

The smaller problems are solved recursively. There are 'a' subproblems at each step.

5. MASTER THEOREM

Compare $f(n)$ with $n^{\log_b a}$
where $\log_b a = \log \text{ base } b \text{ of } a$

Case 1:

If $f(n) < n^{\log_b a}$

Then

$$T(n) = \Theta(n^{\log_b a})$$

Case 2:

If $f(n) = n^{\log_b a}$

Then

$$T(n) = \Theta(n^{\log_b a} \log n)$$

Case 3:

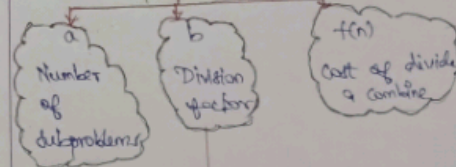
If $f(n) > n^{\log_b a}$
and Regularity condition holds

$$T(n) = \Theta(f(n))$$

Time Complexity

The time complexity is determined using Master Theorem based on the above cases.

4. RECURRENT RELATION
Components



$$T(n) = a T(n/b) + f(n)$$

$a \rightarrow$ Number of subproblems
 $b \rightarrow$ Division Factor
 $f(n) \rightarrow$ Divide & Combine Cost
 $n \rightarrow$ Input Size